

1(a)(i)	concentration / water potential, of sucrose (solution) ;	1
1(a)(ii)	(the coloured sucrose drop / it) would sink / AW ;	1
1(a)(iii)	1 water, leaves potato (by osmosis) ; 2 (sucrose) solution, becomes less dense ;	2
1(b)(i)	(add) 8 cm ³ of stock solution with 12 cm ³ of (distilled) water ;	1
1(b)(ii)	<i>ref. to hazard <u>and</u> risk ;</i>	1
1(b)(iii)	<i>Any six from:</i> 1 use minimum of 5 different concentrations of sucrose solution ; 2 method of maintaining the same temperature (once potato pieces are immersed) ; 3 use same, type / age, of potato ; 4 all pieces of potato are, the same / stated, dimensions ; 5 method to create potato of a certain dimensions ; 6 potato pieces are fully covered / submerged (by sucrose solution) ; 7 remove after same / stated time ; 8 ref. to measuring mass of potato before and after ; 9 method of drying potato pieces (before measuring final mass) ; 10 repeat experiment a minimum of twice and calculate a mean ; 11 <i>idea of</i> repeat with new concentrations of sucrose to get closer to the equivalent sucrose concentration ;	6
1(c)	-1.15 ;	1

Question	Answer	Marks
1(d)(i)	any one from: 1 data is continuous ; 2 data is normally distributed ; 3 comparing (two) means ; 4 standard deviations are approximately the same ; 5 fewer than 30 values / samples (for each potato) ;	1
1(d)(ii)	no difference between, the water potential of, the two types of potato / Record and Kerr's Pink (potato) ;	1
1(d)(iii)	38 ;	1
1(d)(iv)	any three from: 1 critical value (at $df = 38$ and $p = 0.05\%$) = 2.024 ; 2 calculated (t-) value / 1.890 is < than critical (t-) value / 2.024 ; ora 3 null hypothesis is accepted ; 4 there is no <u>significant difference</u> (between the means of the water potentials of the two varieties of potato) ;	3

Question	Answer	Marks
----------	--------	-------